

Cross-Modal Semantic Alignment for Artwork Retrieval

Applied AI Research Seminar / University of Amsterdam

Yuwei Fan(15897273)



A photograph of a museum gallery with green walls and a wooden floor. On the left, a large portrait of a man in 18th-century attire is displayed in an ornate gold frame. To its right, two smaller framed paintings are hung vertically. In the foreground on the left, a marble statue of a nude figure is partially visible. Track lighting is mounted on the ceiling.

Overall

01 Problem & Research Design

02 Methodology & Experimental Setup

03 Empirical Analysis

04 Insights & Conclusions

The background of the slide is a marbled paper pattern. It features swirling, organic shapes in shades of brown, tan, and light blue. A large, solid cream-colored rectangle is centered on the page, serving as a backdrop for the text.

01 Problem & Research Design

Motivation

—WHY SEMANTIC ALIGNMENT IN ART?

- Artworks convey abstract and stylistic meaning
- Textual descriptions often emphasize context, not objects
- Aligning art images and text is more challenging than object images



Title: The Painter's Studio

Artist: Michiel van Musscher

Type: Genre painting

School: Dutch

Timeframe: 1651–1700

Description:

This studio scene depicts a painter gazing directly at the viewer, which can be interpreted as an invitation into his imagined workspace. The composition emphasizes the relationship between the artist, the act of painting, and the observer.

Problem Definition

01

Goal

Align artworks and texts in a shared semantic space

02

Evaluation

Text-to-image retrieval

03

Key challenge

Model choice and textual information

Research Questions

MAIN RQ

How effectively can visual and textual representations be aligned in a shared semantic space for artwork retrieval, and which factors influence the quality of this alignment?

RQ 1

How does the choice of representation model affect the alignment between images and text in artwork retrieval?

RQ 2

How do different forms of textual information, including natural language descriptions and structured metadata, influence retrieval performance under varying levels of difficulty?



Dataset Overall

- **SemArt:** a multimodal dataset of fine-art artworks

- **Each sample includes:**

- 1.an artwork image
- 2.a natural language description
- 3.structured metadata

- **Dataset size (validation set)**

1069 artworks



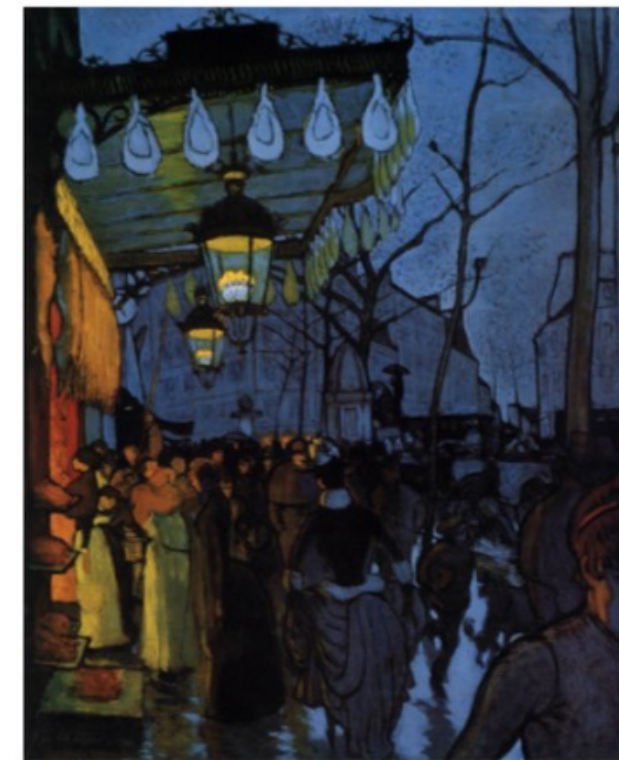
Title: Still-Life
Author: Willem van Aelst
Type: Still-Life
School: Dutch
Timeframe: 1651-1700

The painting depicts a still-life with roses, tulips and other flowers resting on a ledge. It demonstrates the elegance, refinement, and technical brilliance cultivated during the painter's formative years in Italy.



Title: Grape Harvest Girl
Author: Ljubomir Aleksandrova
Type: Genre
School: Other
Timeframe: 1851-1900

In Croatia, Bosnia and Herzegovina, and in northern Serbia, depending on the kind of harvest, people celebrate harvest season by dressing themselves with fruits of the harvest.



Title: Avenue de Clichy - Five O'Clock in the Evening
Author: Louis Anquetin
Type: Landscape
School: French
Timeframe: 1851-1900

This painting is said to have inspired Van Gogh in painting his famous Café Terrace at Night.

Figure adapted from Garcia and Vogiatzis (2018).



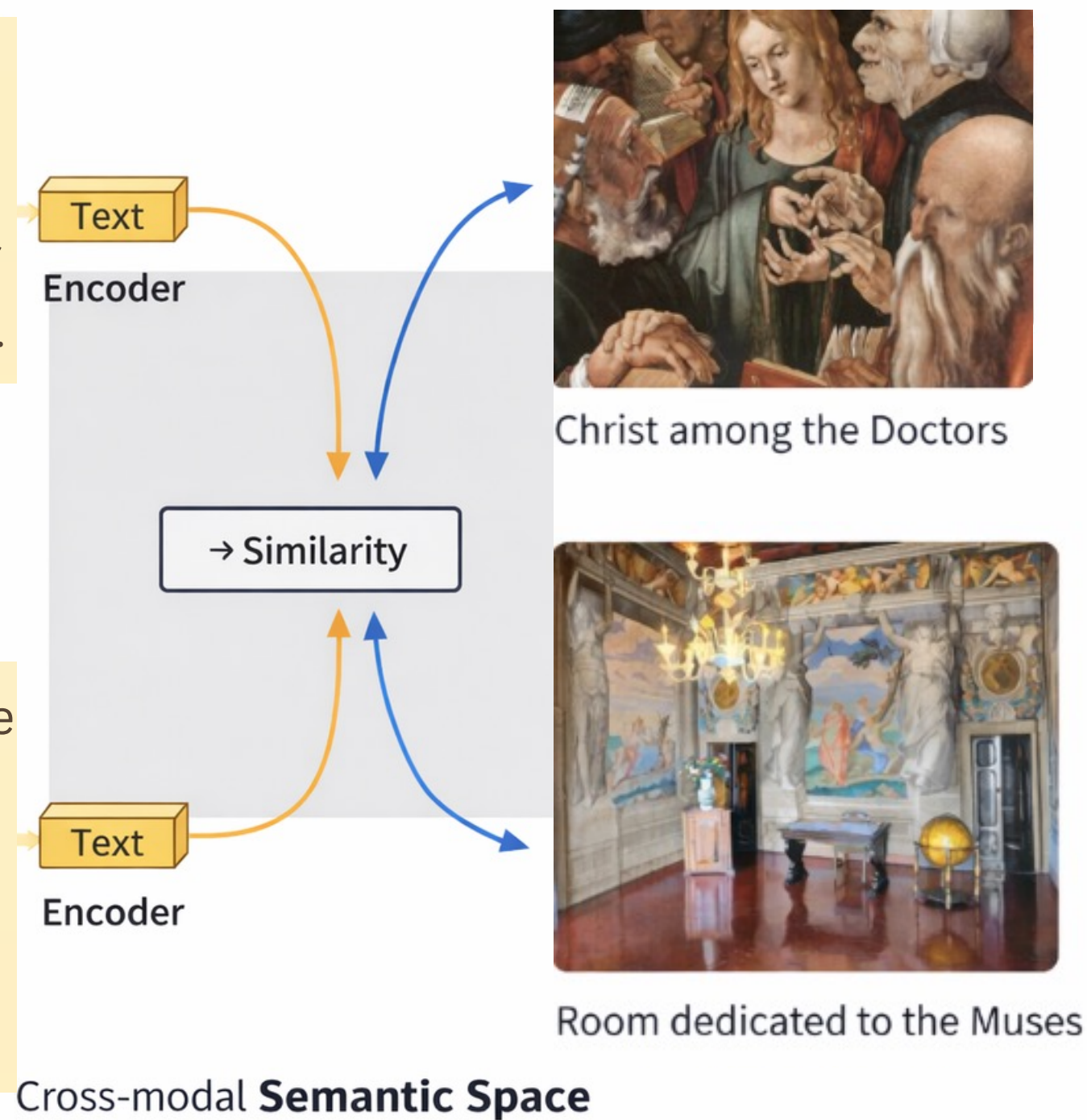
02

Methodology & Experimental Setup

Methodology+ Textual Queries

At the same time as the Feast of the Rose Garlands, Dürer was working on the painting of Christ among the Doctors.

The room dedicated to the Muses is highly theatrical. The ceiling seems to be held up by gigantic caryatids that appear on the walls.



Methodology Overview

- Cross-modal semantic alignment evaluated via retrieval
- Images and texts mapped into a shared embedding space
- Ranking based on cosine similarity
- Recall@K and MRR as evaluation metrics

Query Construction

- **Desc:** natural language descriptions
- **Meta:** structured metadata (artist, school, period...)
- **Desc + Meta:** combined queries
- **Structured:** structured prompt with same info

Representation Models



Cross-Modal Models



Unimodal baseline (sanity check)

01
CLIP ViT-B/32

Joint image-text embedding

ResNet-50

image-only

02
CLIP RN50

Joint image-text embedding

Experimental Setup

Query



10 image
candidates
(ranked list)

```
===== EASY =====
idx_easy=2 | GT cand#2 | GT rank#1
Query (first 350 chars):
Isack van Ostade was the short-lived younger brother of the painter of scenes from peasant life, Adriaen van Ostade. They were both born, lived and worked in Haarlem, and Isack was a
pupil of his brother who was eleven years his senior. Isack joined the guild in 1643. His earliest paintings - the first dated picture is from 1639 - are peasant inter ...

===== DIFFICULT =====
idx_diff=2 | GT cand#10 | GT rank#6
Query (first 350 chars):
The subject of the scene is an illustration of the popular adage "Sine Baccho et Cerere friget Venus" (Without Ceres and Bacchus, Venus would Freeze): without food (Ceres, the Roman
god of agriculture) and wine (Bacchus), love (Venus) is left cold ...
```



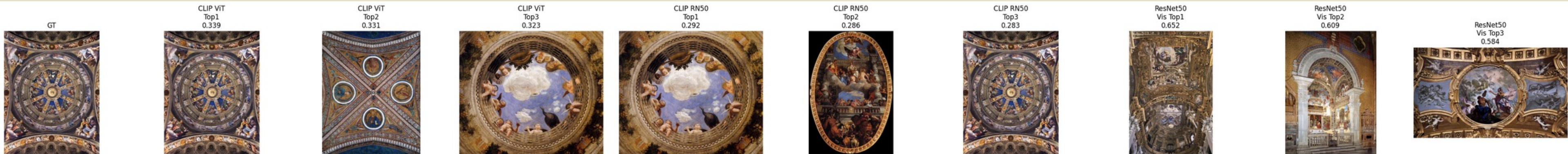
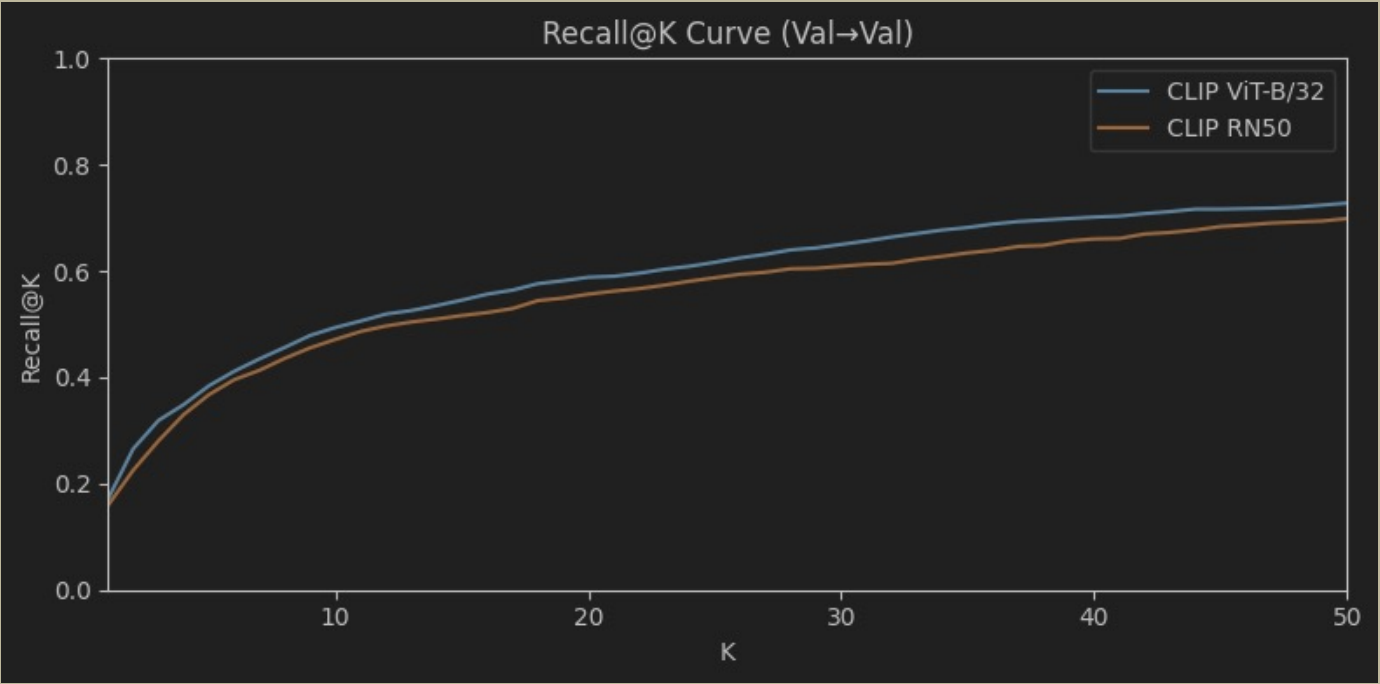
03 Empirical Analysis

Experiment 1: Model Selection



- CLIP ViT-B/32 achieves higher Recall@K and MRR than CLIP RN50 across all metrics
- This result motivates the selection of ViT-B/32 for subsequent experiments
- ResNet-50 is included as a vision-only baseline for sanity checking the retrieval pipeline.

Model	R@1	R@5	R@10	MRR
CLIP ViT-B/32	0.171	0.384	0.494	0.279
CLIP RN50	0.159	0.368	0.471	0.257



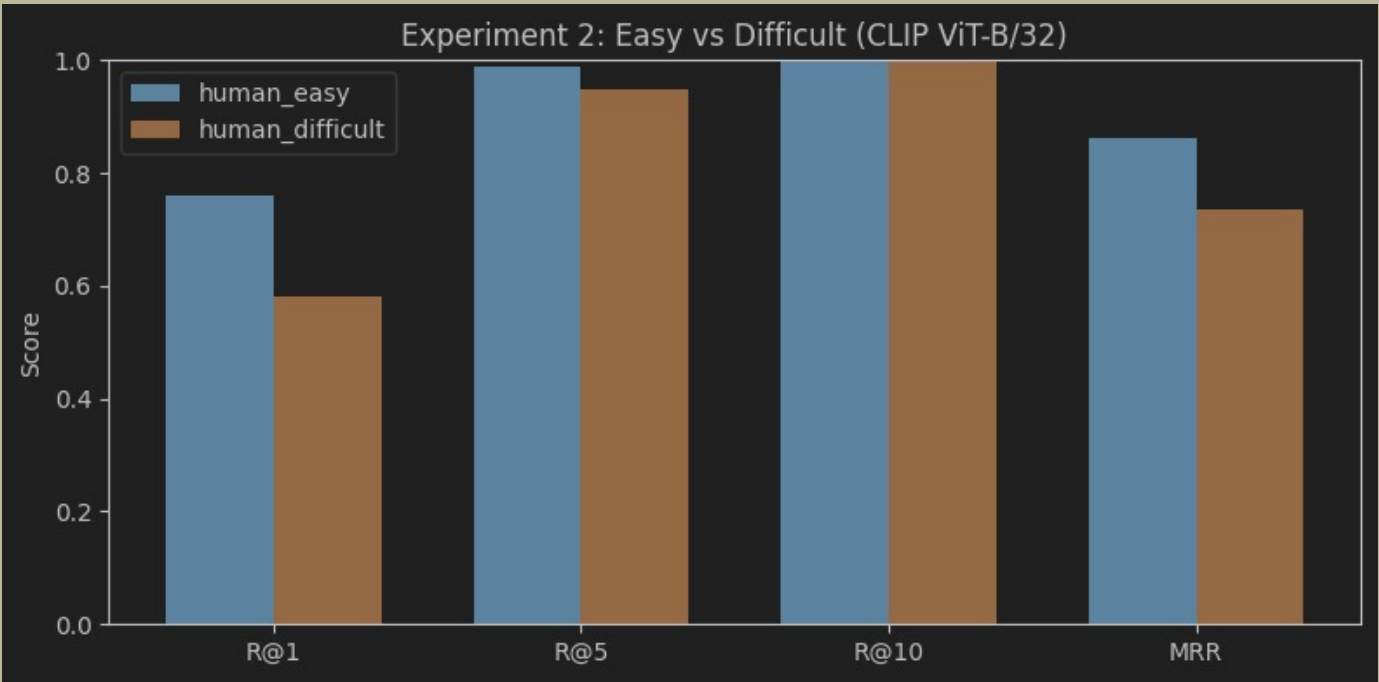
Note: For ResNet-50, the Top-3 results exclude the query image itself

Experiment 2:

Textual Information and Retrieval Difficulty

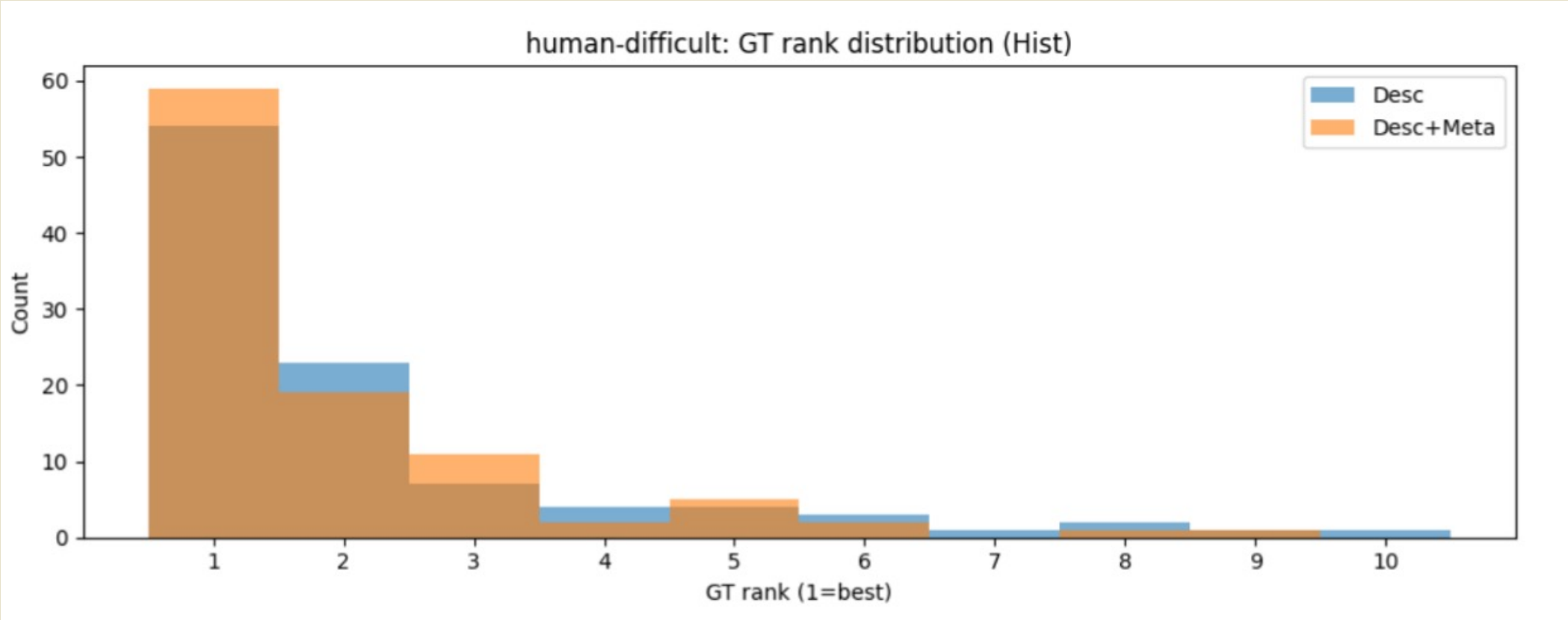
- Fixed model: CLIP ViT-B/32
- Retrieval task: Text → Image
- Text query types:
 - Description (Desc)
 - Metadata (Meta)
 - Description + Metadata (Desc+Meta)
 - Structured (a structured prompt)
- Difficulty split:
 - Easy subset
 - Human-defined Difficult subset

Setting	Desc	Meta	Desc+Meta
Easy	strong	weak	similar
Difficult	strong	weak	improved

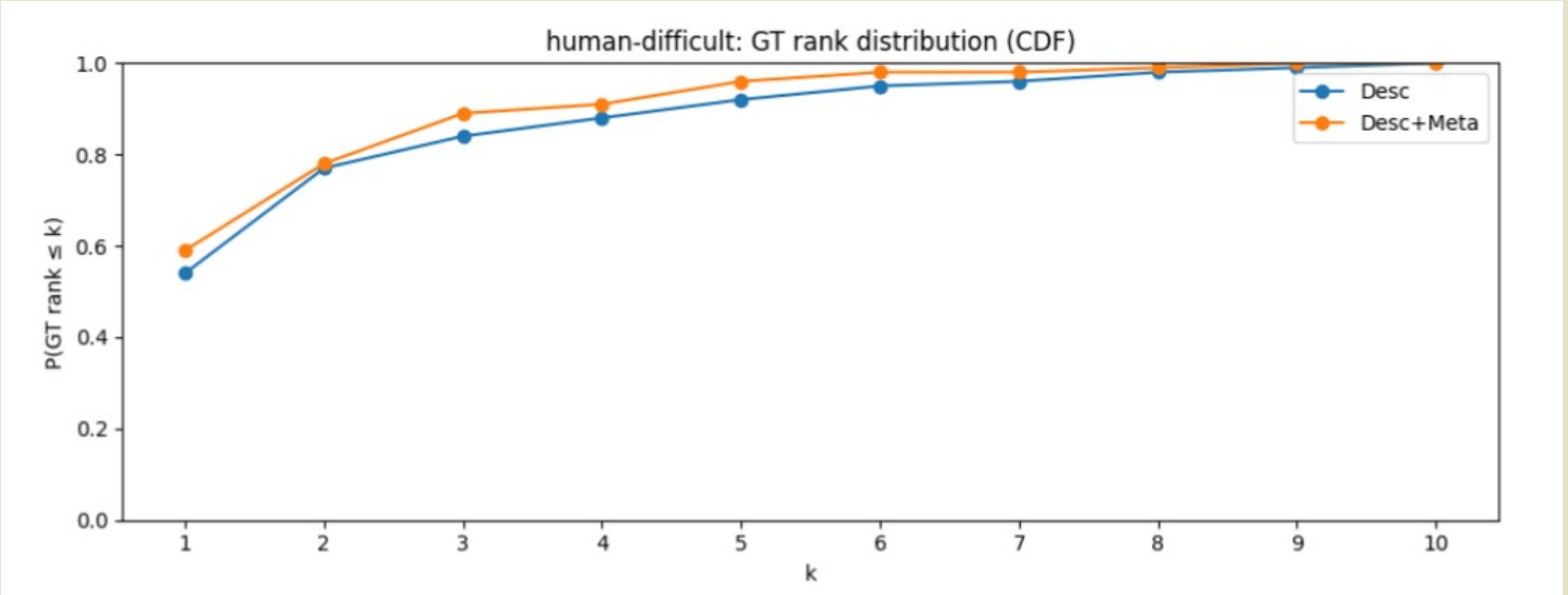


Split	Query	R@1	R@5	R@10	MRR
Easy	Desc	0.74	0.98	1.00	0.840
	Meta	0.45	0.94	1.00	0.648
	Desc+Meta	0.76	1.00	1.00	0.857
	Structured	0.80	0.99	1.00	0.886
Difficult	Desc	0.54	0.92	1.00	0.707
	Meta	0.22	0.70	1.00	0.436
	Desc+Meta	0.59	0.96	1.00	0.742
	Structured	0.56	0.95	1.00	0.713

GT Rank Distribution under Difficult Retrieval



GT rank histogram (Desc vs Desc+Meta)



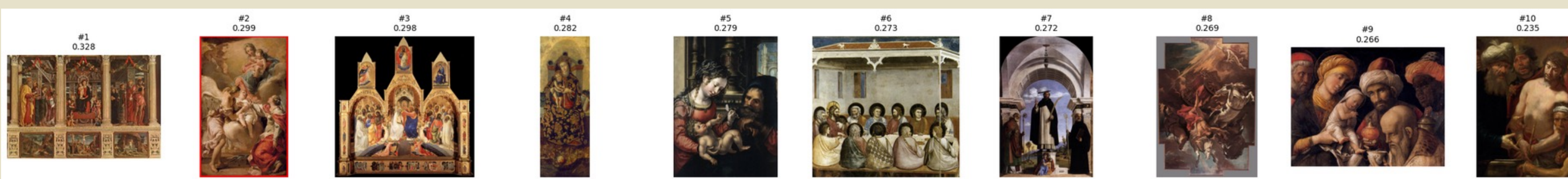
GT rank CDF (Desc vs Desc+Meta)

Improved cases

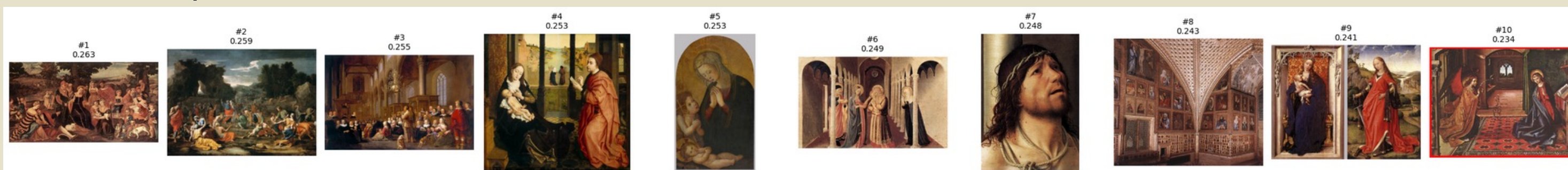
mode=Desc | GT_rank=8



mode=Desc+Meta |GT_rank=2



mode=Desc | GT_rank=10



mode=Desc+Meta |GT_rank=6



No-improvement cases

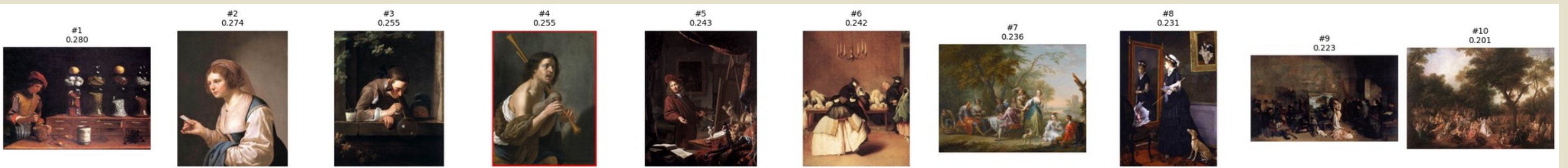
mode=Desc | GT_rank=6



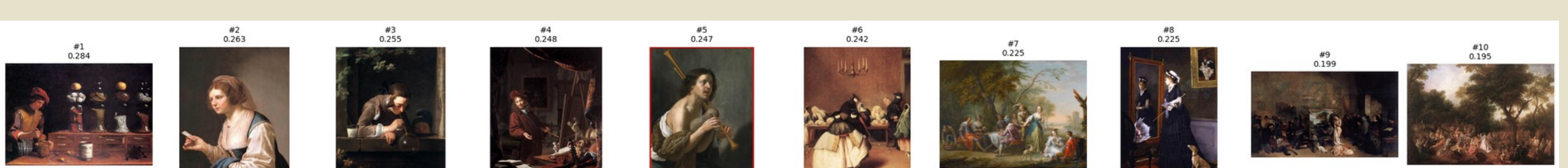
mode=Desc+Meta |GT_rank=8



mode=Desc | GT_rank=4



mode=Desc+Meta |GT_rank=5





04

Insights & Conclusions

Key Insight

1. Representation-Level Insights

- CLIP ViT-B/32 provides stronger semantic alignment than CLIP RN50
- Architectural choice significantly affects cross-modal retrieval performance

2. Textual Information Insights

- Natural language descriptions capture rich semantic content
- Metadata alone lacks fine-grained visual semantics
- Metadata complements descriptions in ambiguous retrieval cases

3. Evaluation Insights

- Difficulty-aware evaluation reveals effects hidden in average metrics
- Qualitative cases show metadata can both help and introduce noise

Conclusion and Future Work

- This study investigates cross-modal semantic alignment for artwork retrieval
- Retrieval-based evaluation provides a practical proxy for alignment quality
- Results highlight the importance of model choice and textual representation

Conclusion

- Scale to larger candidate pools
- Fine-tune models on art-specific data
- Incorporate richer structured knowledge

Future Work



THANK YOU